System Design Report (Milestone 3)

CSC-225 – Software Engineering

Department of Computer Science

Namal University, Mianwali

Project Title: Namal Criko

Cricket Simulation Trainer

Team Members:

Name	Roll No.	Email
M. Jamal Ahmad Khan	Num-Bscs-2024-51	bscs24f51@namal.edu.pk
Qazi M. Auon Farooqi	Num-Bscs-2024-64	bscs24f64@namal.edu.pk
Breera Ijaz	Num-Bscs-2024-20	bscs24f20@namal.edu.pk

Instructor: Asiya Batool

Requirement Provider: Mr. Ammar Ahmad Khan

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Contents

1 Introduction

Namal Criko is a web-based cricket simulation training platform designed for Namal University students. The system provides interactive simulations of cricket techniques (batting and bowling) enabling students to learn and practice in a virtual environment.

This Design Report translates the approved Software Requirements Specification (SRS) into a complete system design including behavioral diagrams, structural diagrams, system architecture, prototypes, and traceability documentation.

2 Design Assumptions and Constraints

2.1 Design Assumptions

1. Users have basic web browsing skills and cricket knowledge
2. Admin approval occurs within 24-48 hours
3. Users provide accurate registration information
4. University network provides adequate bandwidth
5. Administrative staff available for user approval
6. Cricket experts validate technique accuracy
7. University continues post-development support

2.2 Design Constraints

Technology: Limited to open-source frameworks
Budget: Zero-budget project
Team: Three members with limited experience
Timeline: One academic year
Hosting: Free-tier cloud services
Hardware: Must work on student devices
Software: No proprietary integrations
Operational: Limited post-semester maintenance

3 Key Design Decisions

3.1 DFD Decomposition

Level 0: System boundary with external entities (Player, Admin, Email Service)
Level 1: Six major processes (User registration and login, Admin Controller, Simulation Manager, Help and ticket manager, Notification and Event Manager, Report and analysis)
Level 2: Detailed decomposition of complex processes

Justification: Follows functional logic; single responsibility per process

3.2 Class Relationships

Inheritance: *Player* and *Coach* extend the base *User* class to share authentication and profile attributes while maintaining specific roles.

Association: *Player* has a one-to-many relationship with *TrainingSession*; a player can perform multiple drills over time.

Composition: *Simulation* consists of *Frames* (e.g., Stance, Impact, Follow-through). If a simulation is deleted, these specific 3D frames are also destroyed.

Aggregation: *CoachingManual* aggregates *TechniqueGuides*. The guides exist independently of the manual and can be updated or reorganized.

3.3 Justification for Class Relationships

- **Inheritance** was chosen to simplify the User Management module, ensuring that both Coaches and Players follow the same security protocols while having different dashboard access.
- **Composition** for the Simulation class ensures data integrity; since a "Frame" (like the Forward Defense stance) has no meaning outside the context of a training session, it is strictly bound to its parent object.
- **Separation of Concerns:** By decoupling the *AIAnalysisEngine* from the *UIController*, we ensure the 3D simulation remains performant even during heavy data processing.

3.4 System Architecture

Architecture: Three-tier (Presentation-Application-Data)

- **Presentation (Frontend):** Developed using modern web technologies to render the 3D Cricket Simulation and the "Coaching Manual" dashboard.
- **Application (Logic):** This layer processes simulations frames to present techniques like the "Forward Defense" as per ideal standards.
- **Data (Storage):** A secure database layer that manages player profiles, training history, and the library of high-fidelity "Ideal Technique" reference frames.

UI: Desktop Familiar design

4 System Modeling Diagrams

4.1 Use Case Diagram

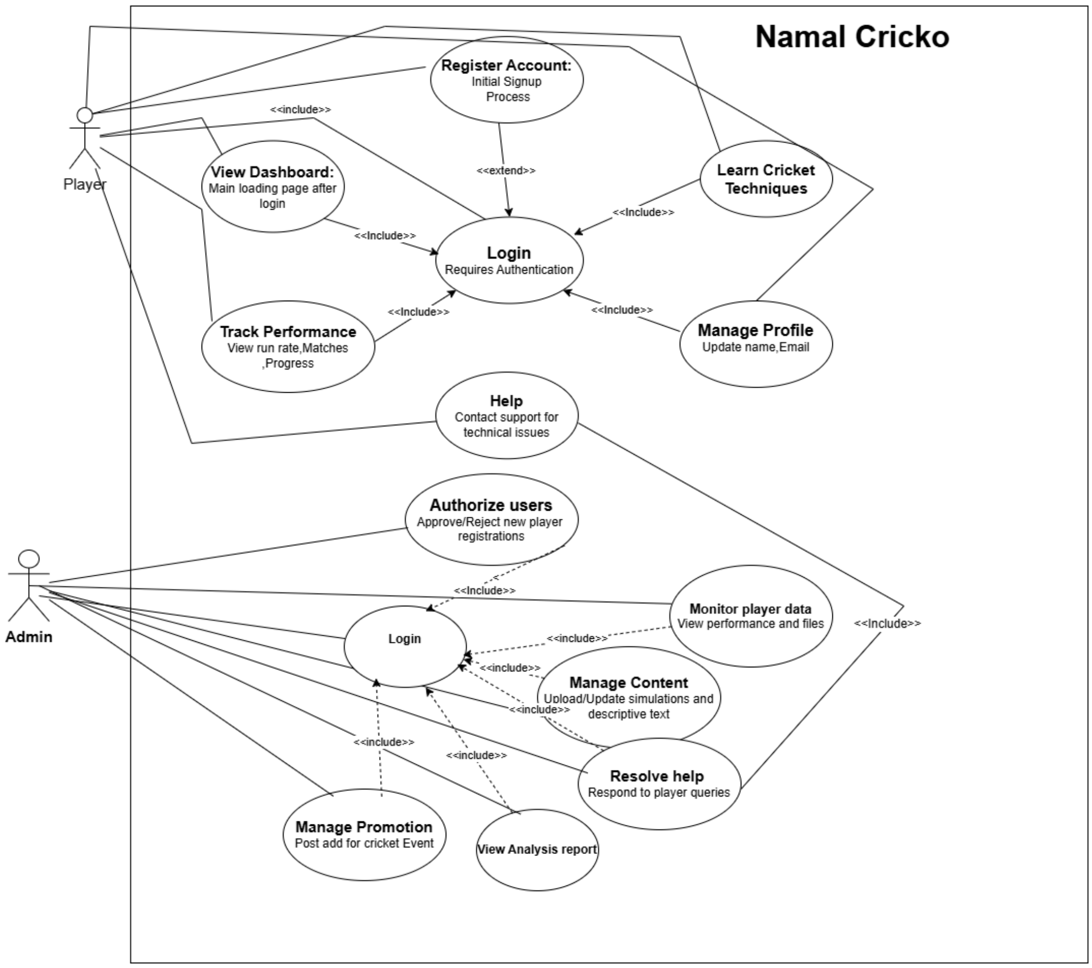


Figure 1: Use Case Diagram

4.2 Data Flow Diagrams

4.2.1 DFD Level 0

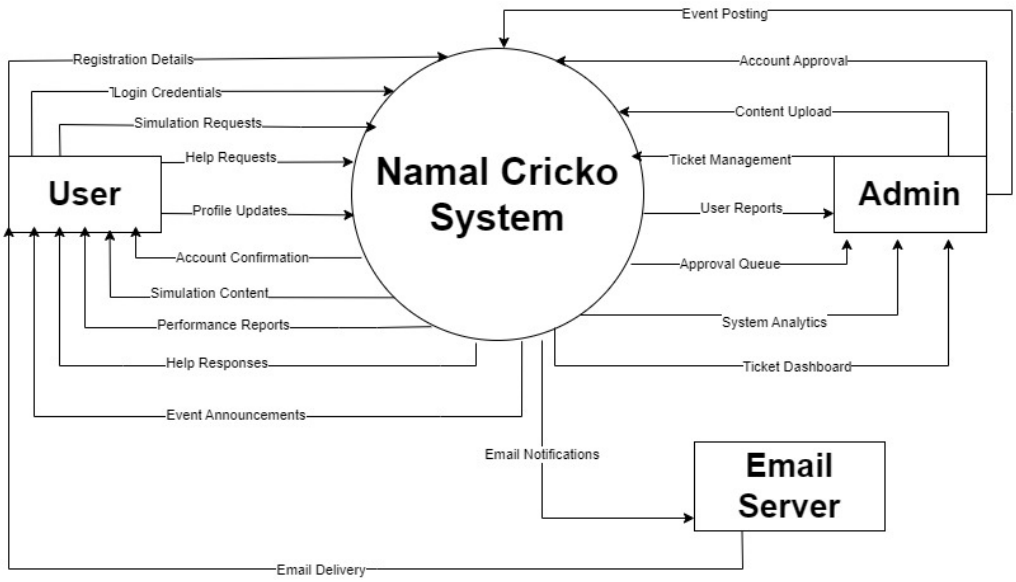


Figure 2: DFD Level 0 - Context Diagram

4.2.2 DFD Level 1

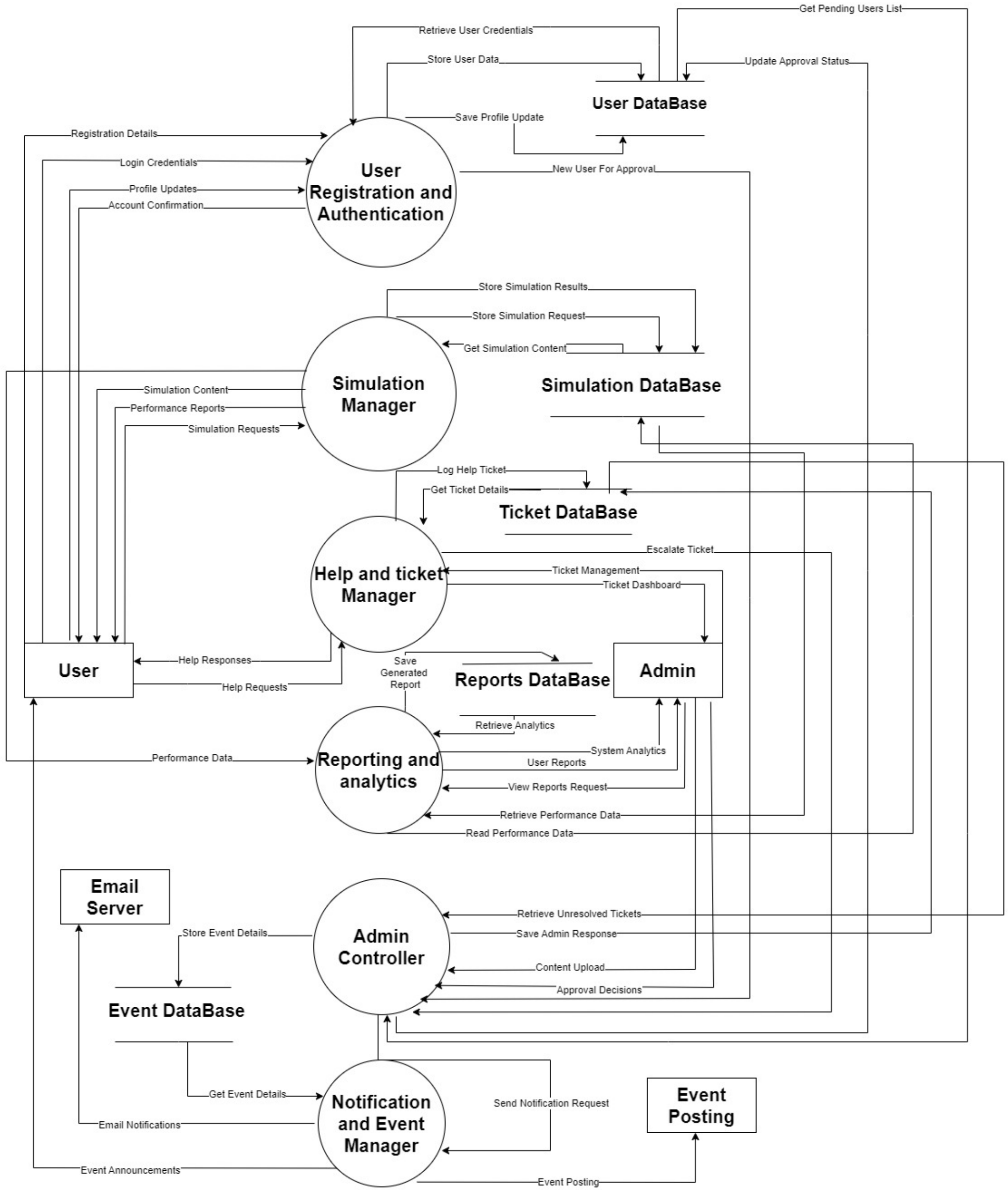


Figure 3: DFD Level 1 Diagram

4.2.3 DFD Level 2

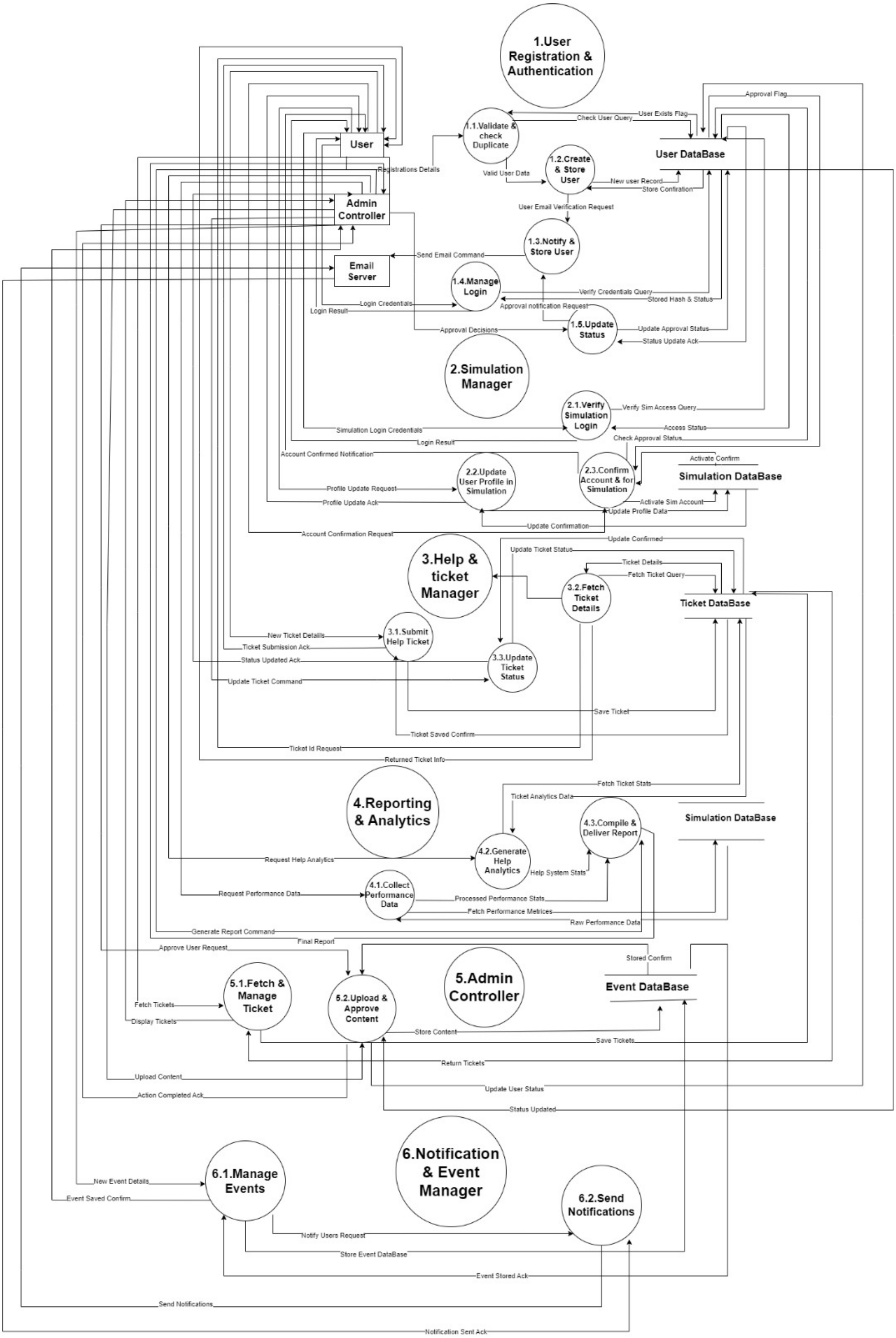


Figure 4: DFD Level 2

4.3 Sequence Diagrams

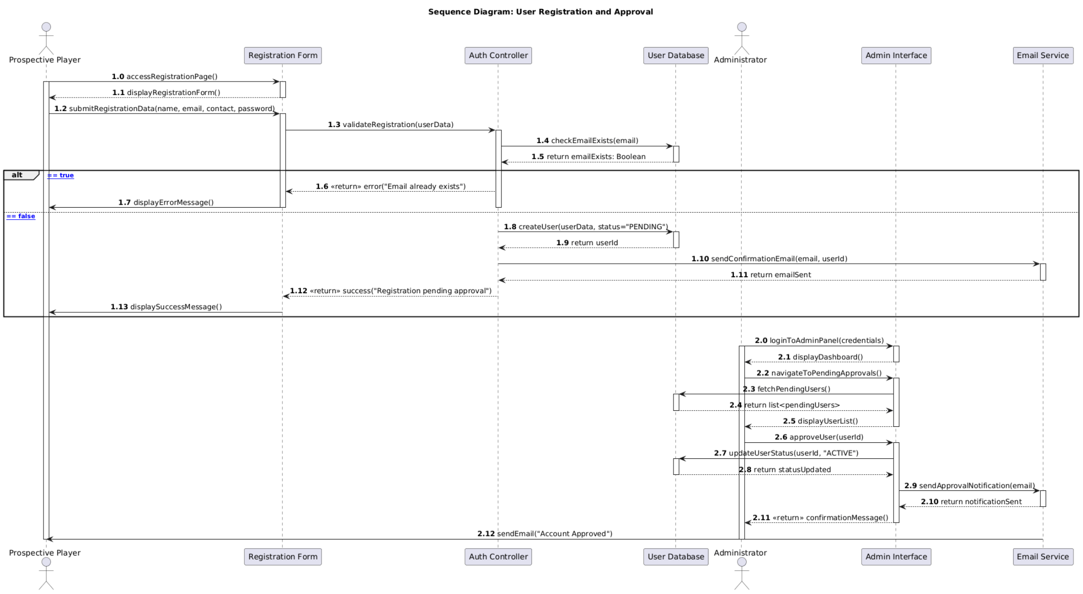


Figure 5: SD-01: User Registration

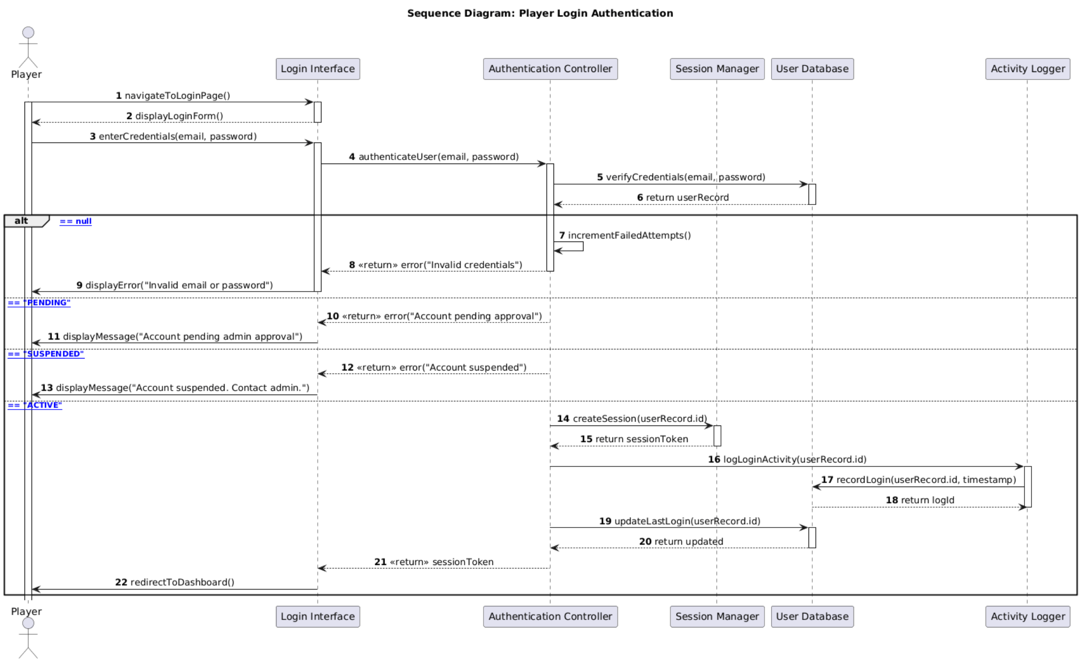


Figure 6: SD-02: User Login

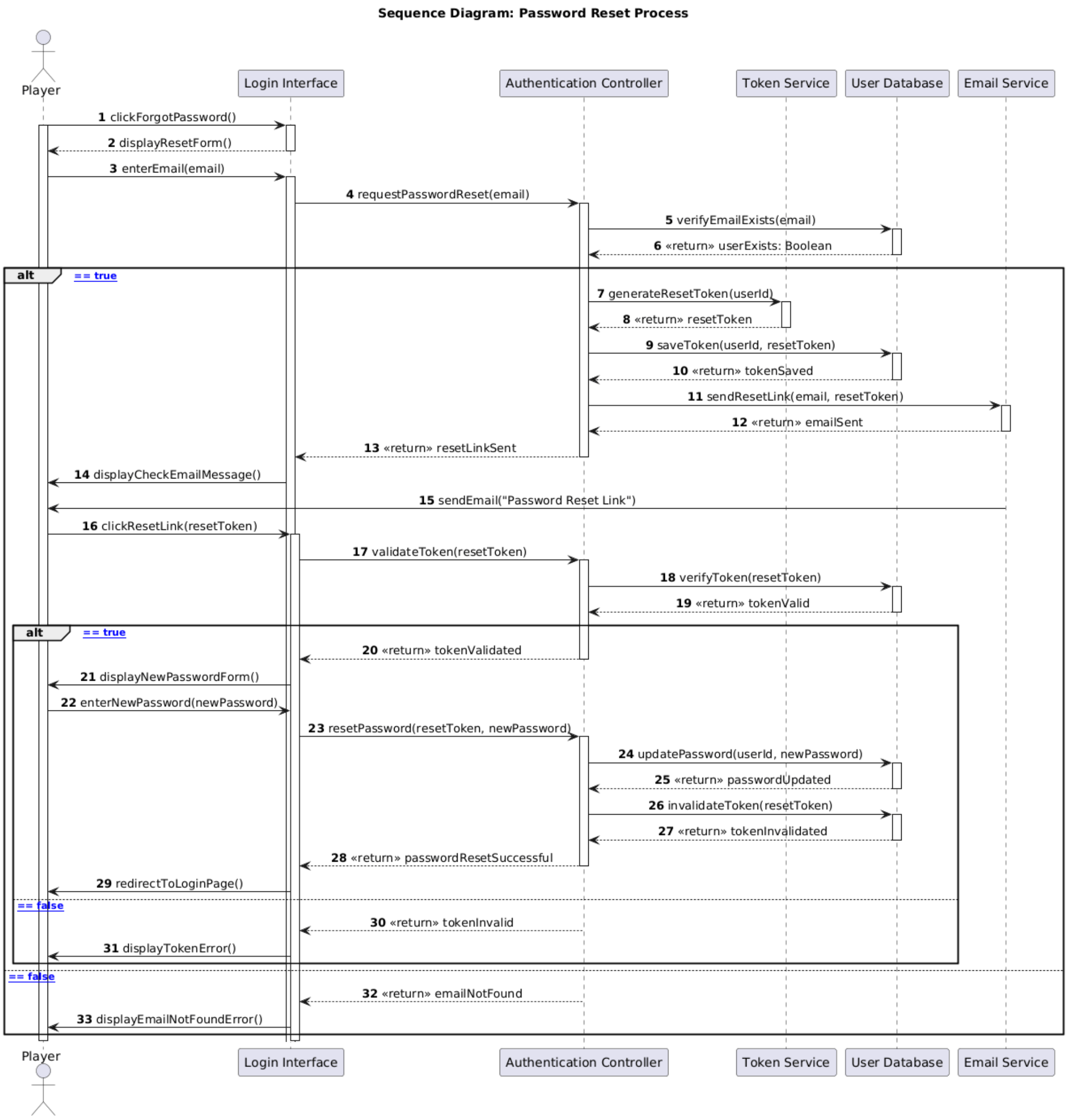


Figure 7: SD-03: Password Reset

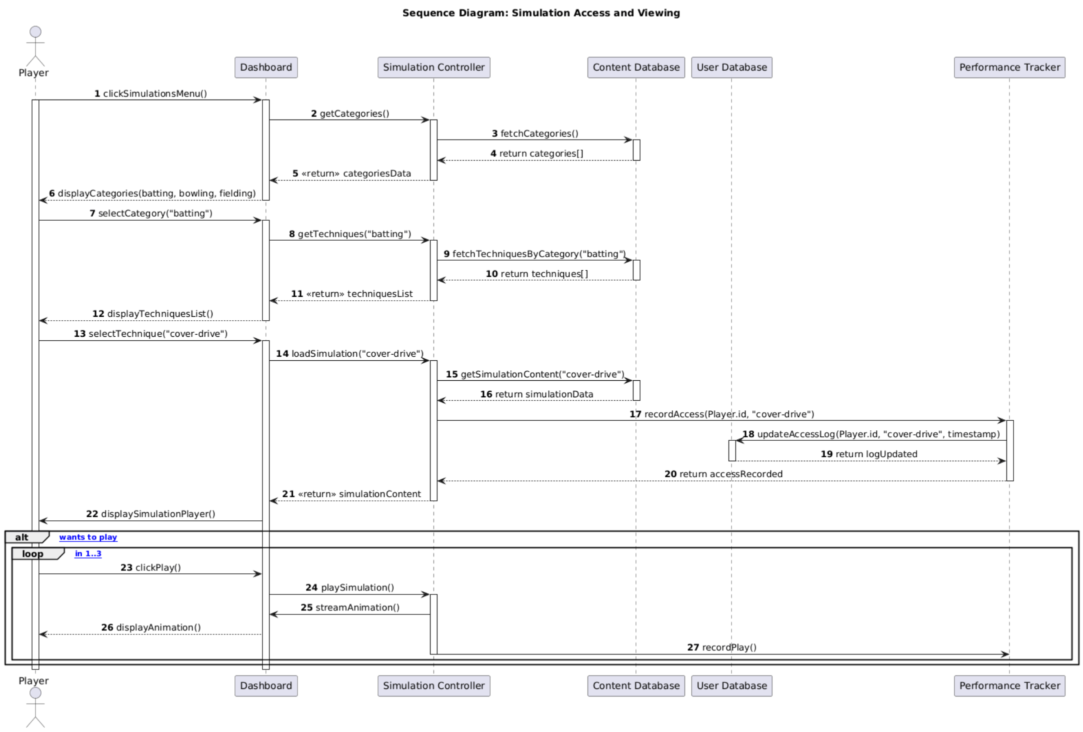


Figure 8: SD-04: Simulation Viewing

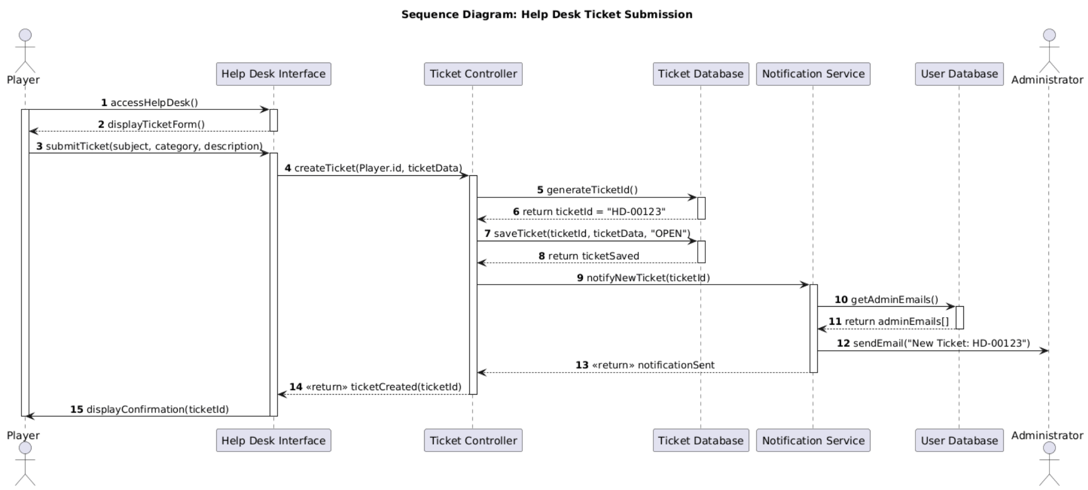


Figure 9: SD-05: Help Desk Ticket Submission

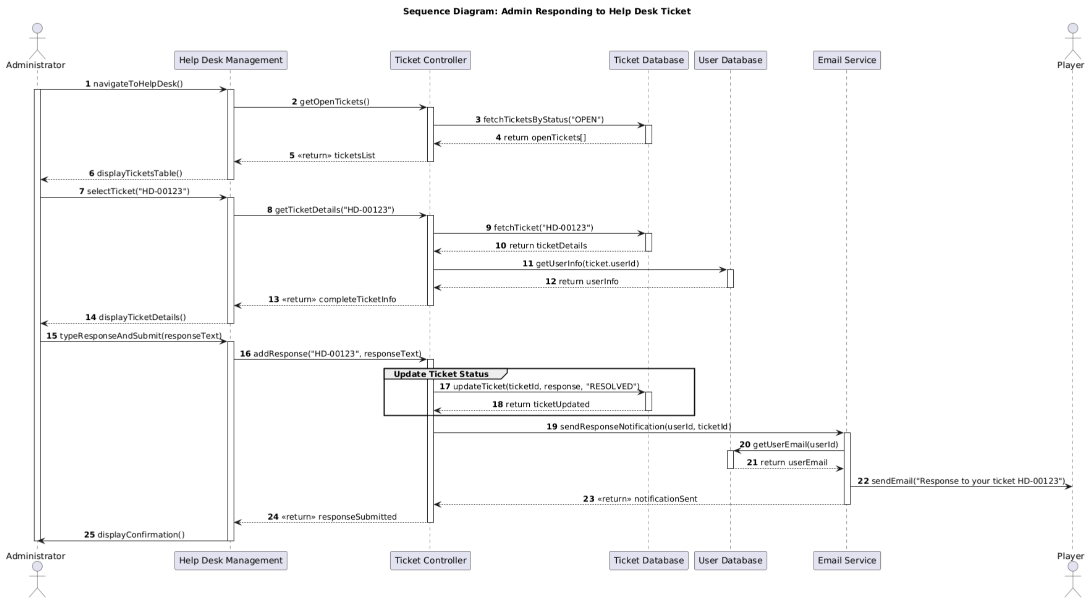


Figure 10: SD-06: Admin Ticket Response

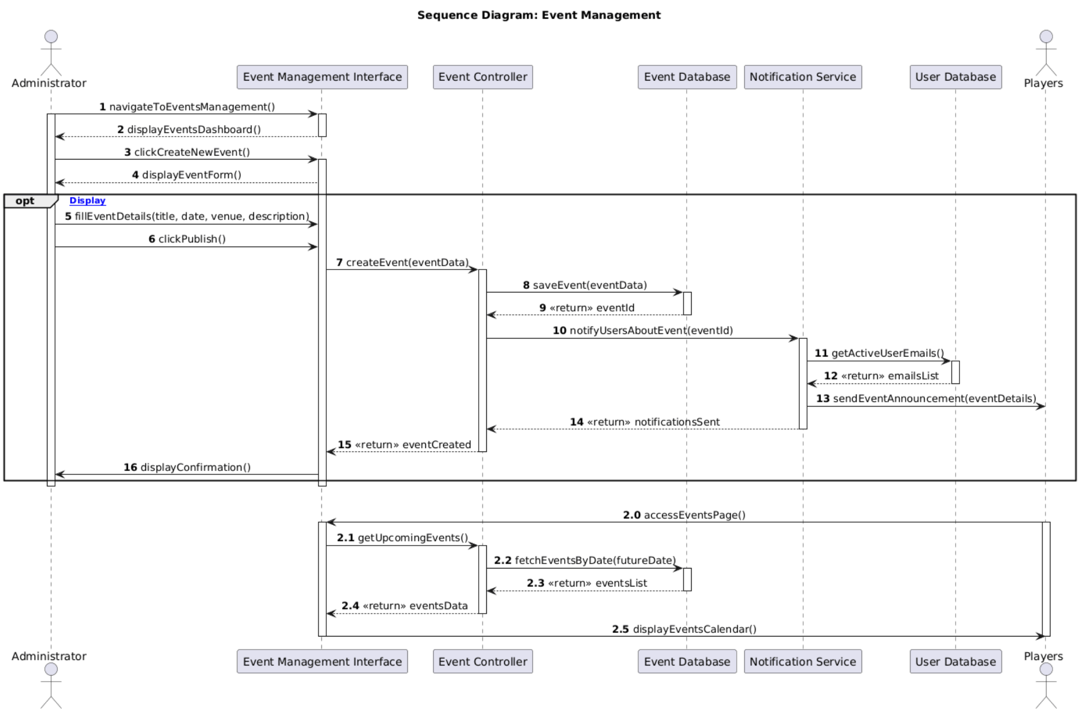


Figure 11: SD-07: Event Management

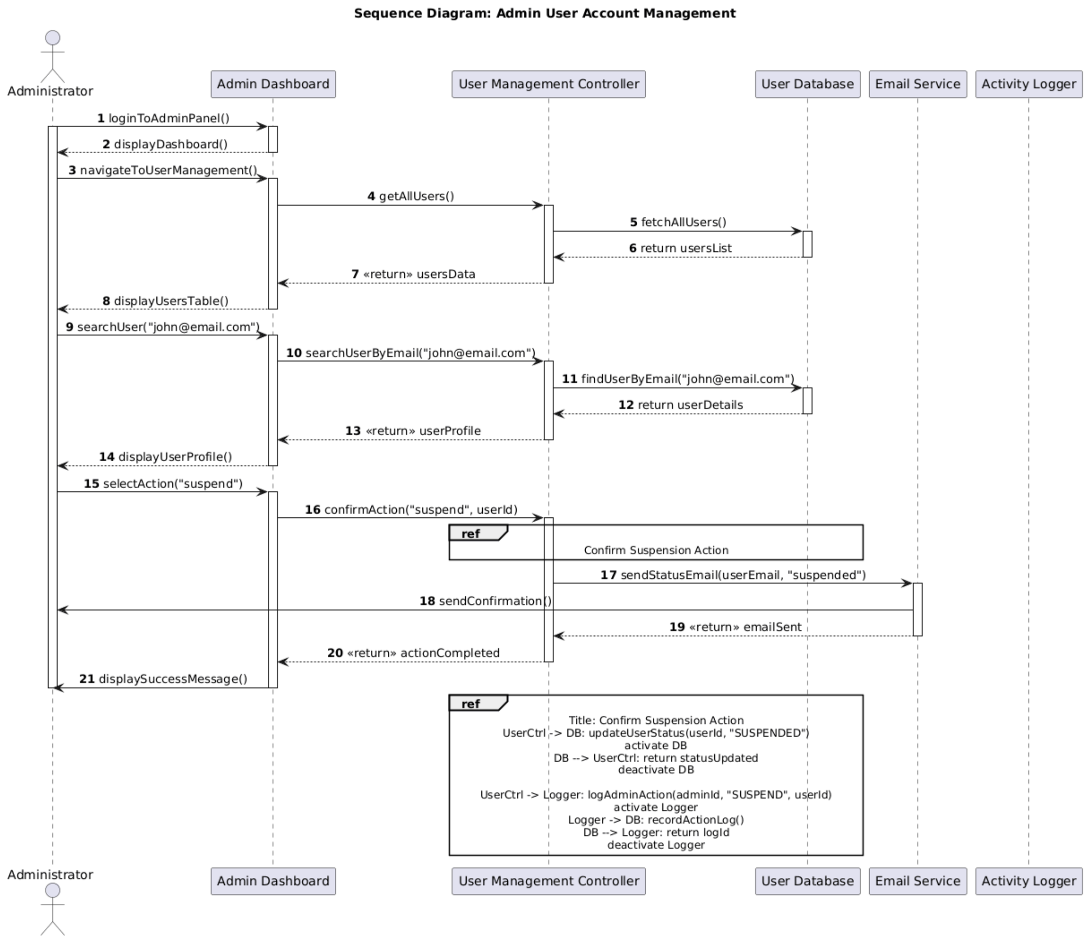


Figure 12: SD-08: Admin User Management

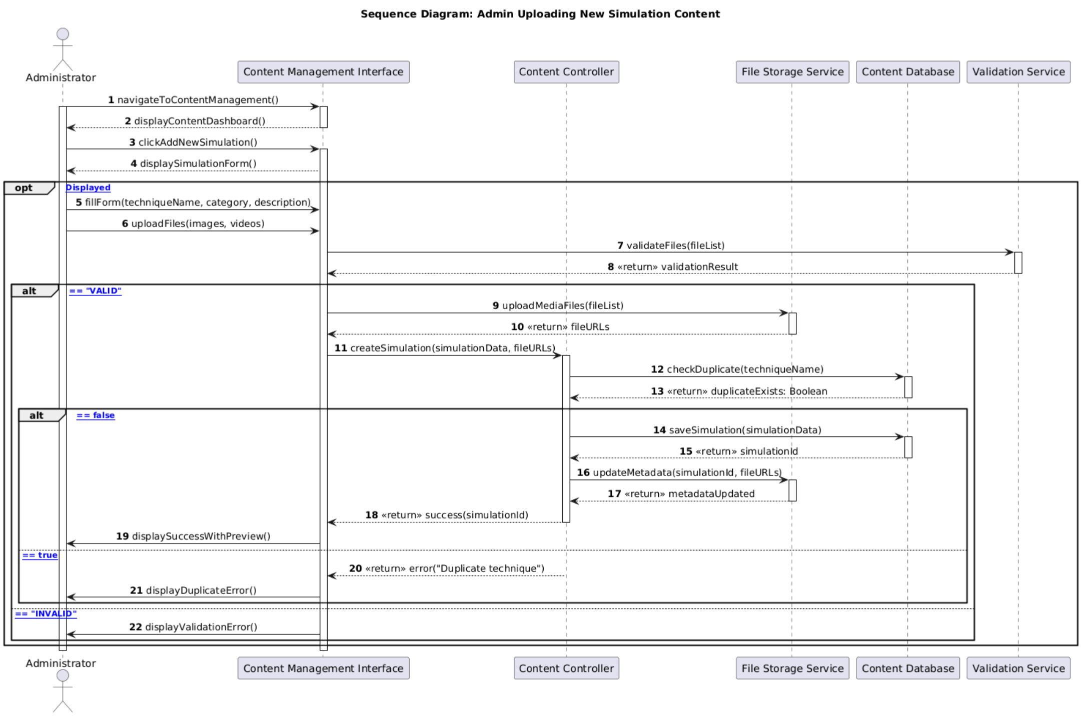


Figure 13: SD-09: Simulation Management

4.4 Activity Diagrams

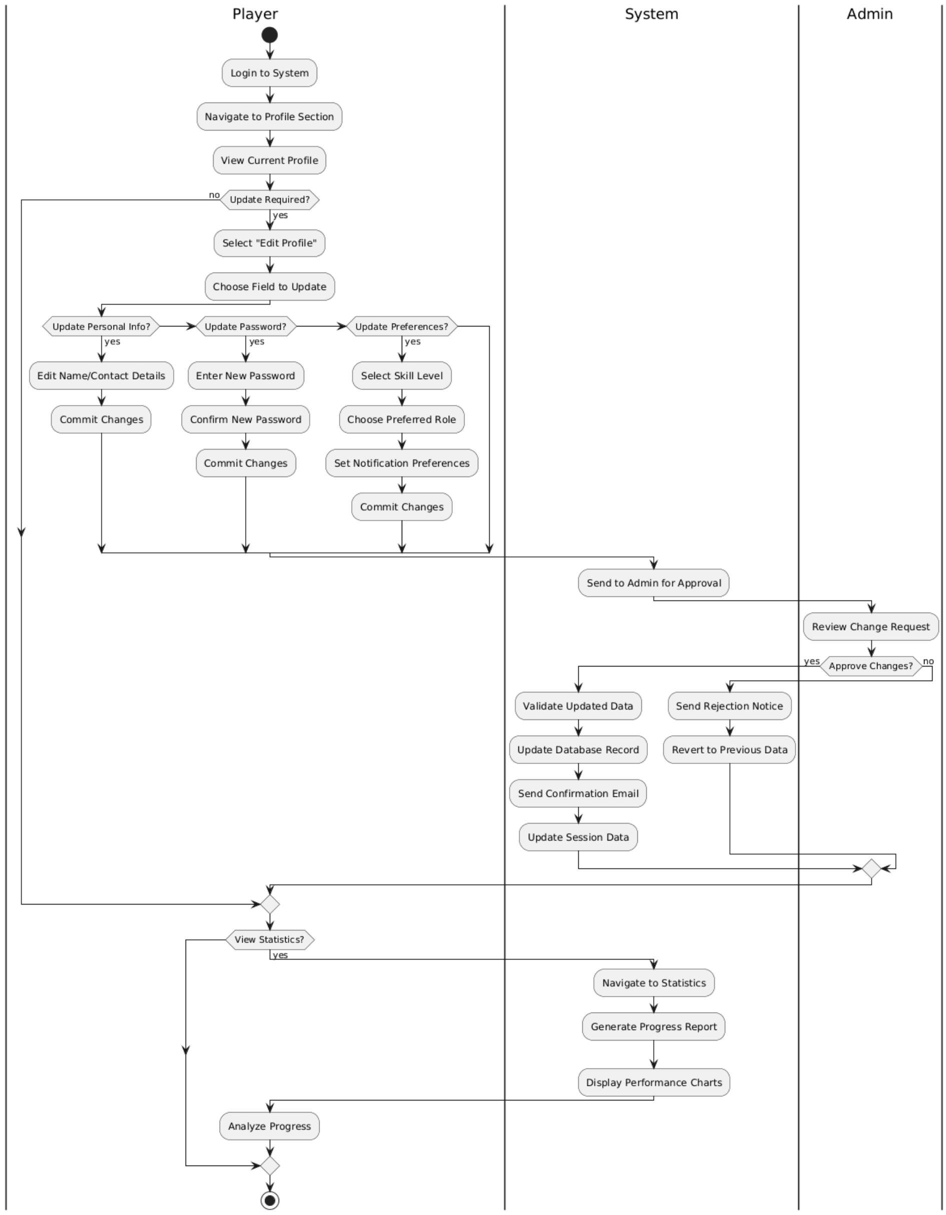
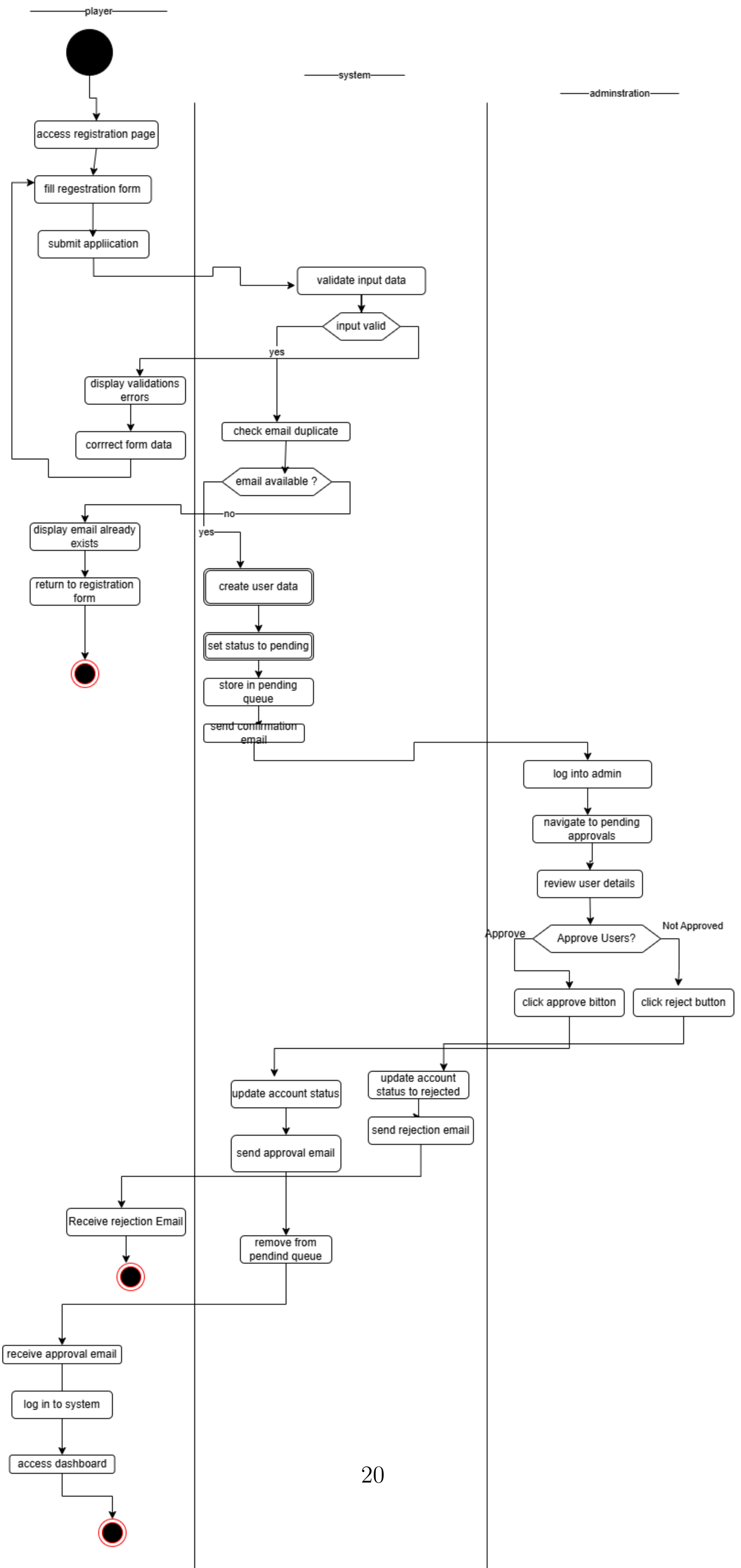


Figure 14: ACT-01: Profile Management


player
system
adminstration
access registration page
fill regestration form
submit appliication
validate input data
input valid
yes
display validations errors
corrrect form data
check email duplicate
email available ?
no
yes
display email already exists
return to registration form
create user data
set status to pending
store in pending queue
send confirmation email
log into admin
navigate to pending approvals
review user details
Approve
Approve Users?
Not Approved
click approve bitton
click reject button
update account status
update account status to rejected
send rejection email
send approval email
Receive rejection Email
remove from pendind queue
receive approval email
log in to system
access dashboard

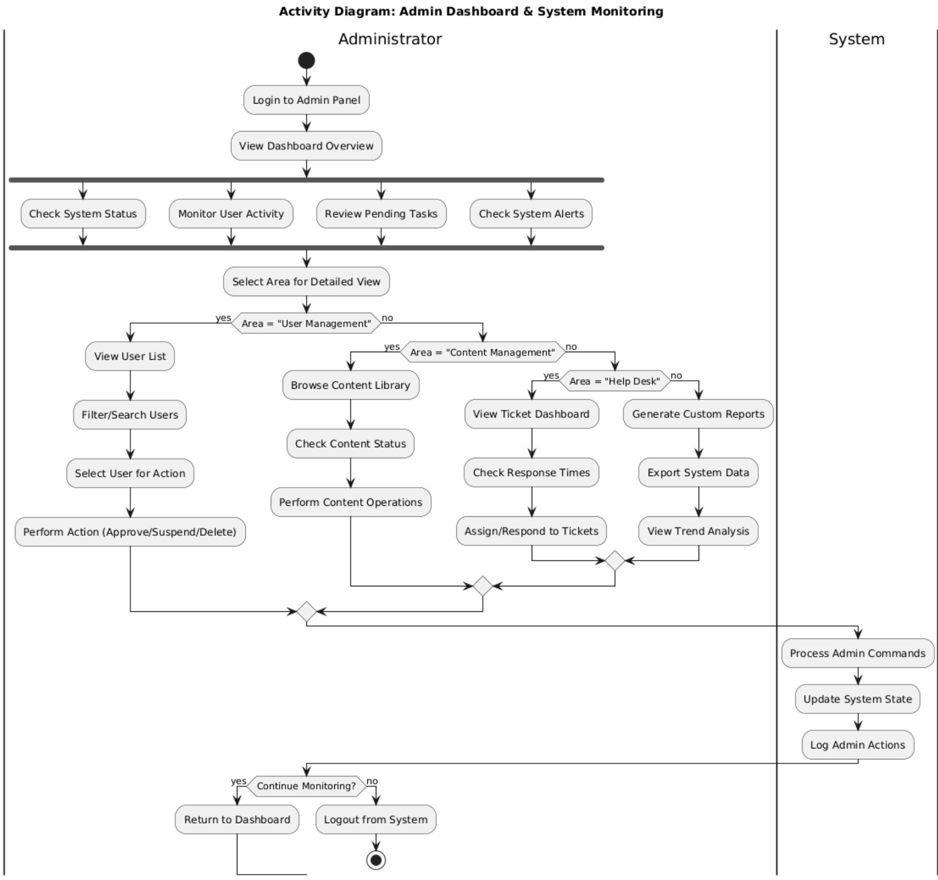


Figure 16: ACT-03: Admin Dashboard

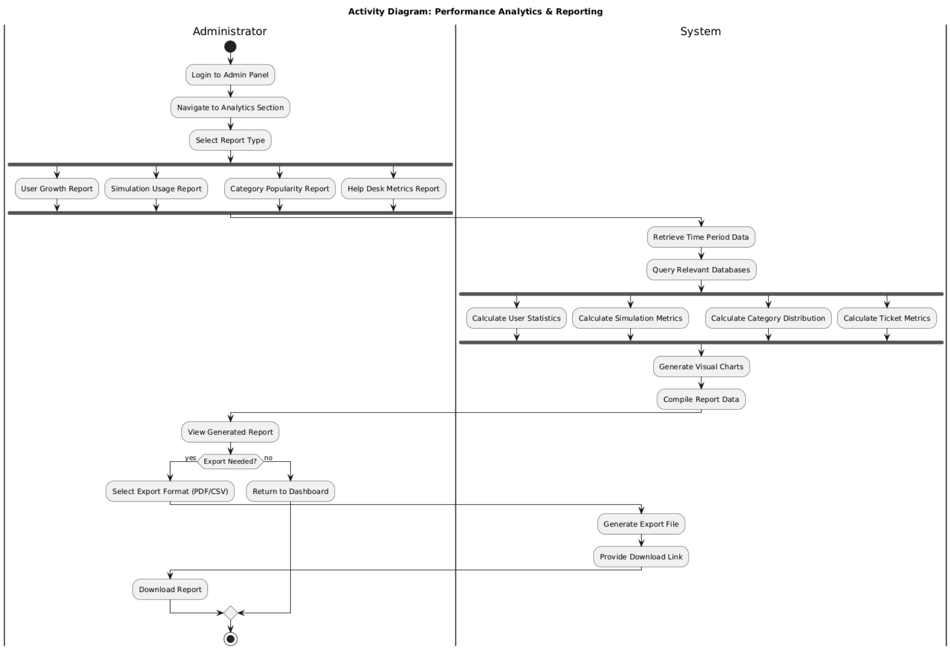


Figure 17: ACT-04: Analytics Report Generation

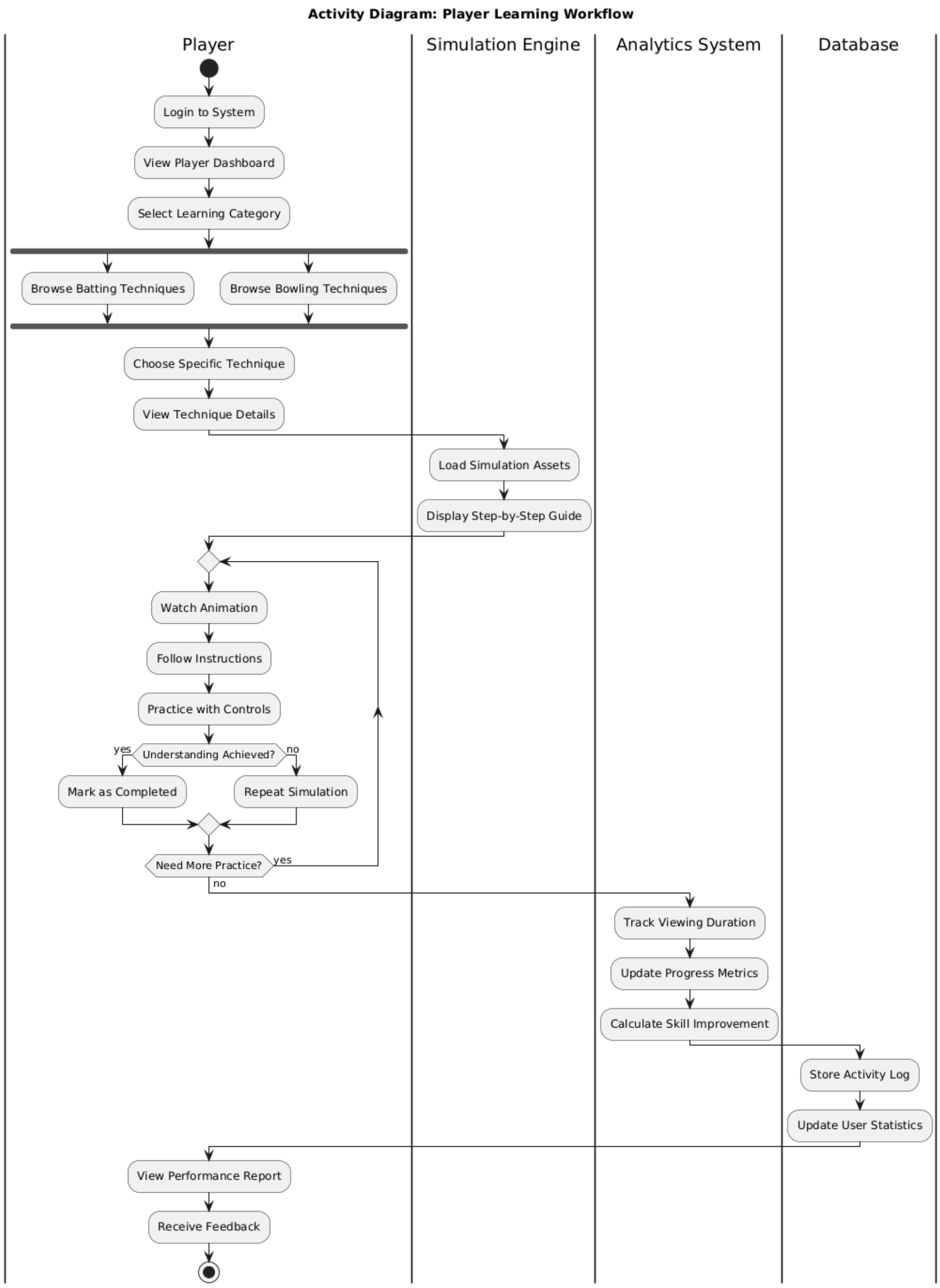


Figure 18: ACT-05: Simulation Navigation

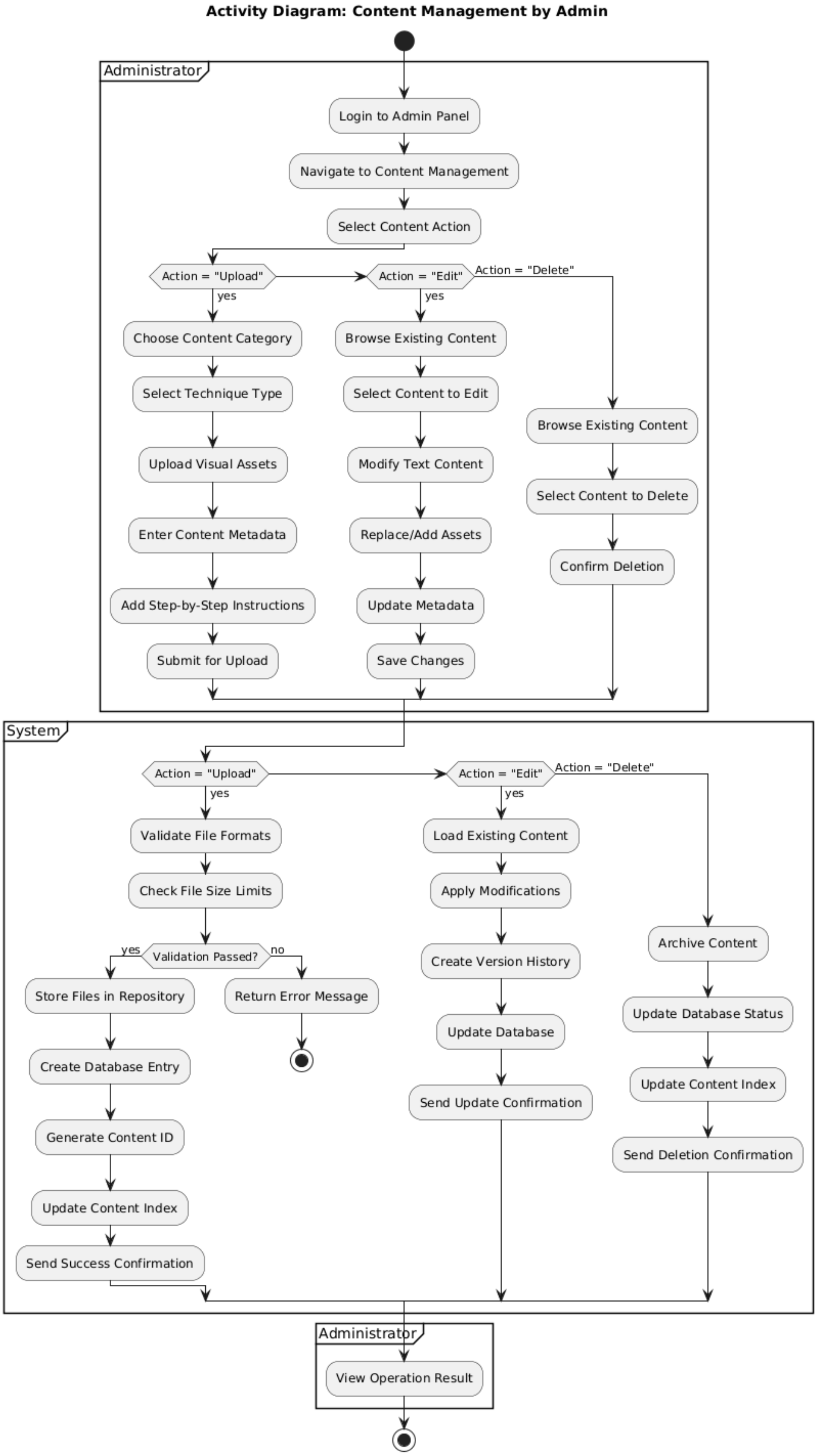


Figure 19: ACT-06: Content Management

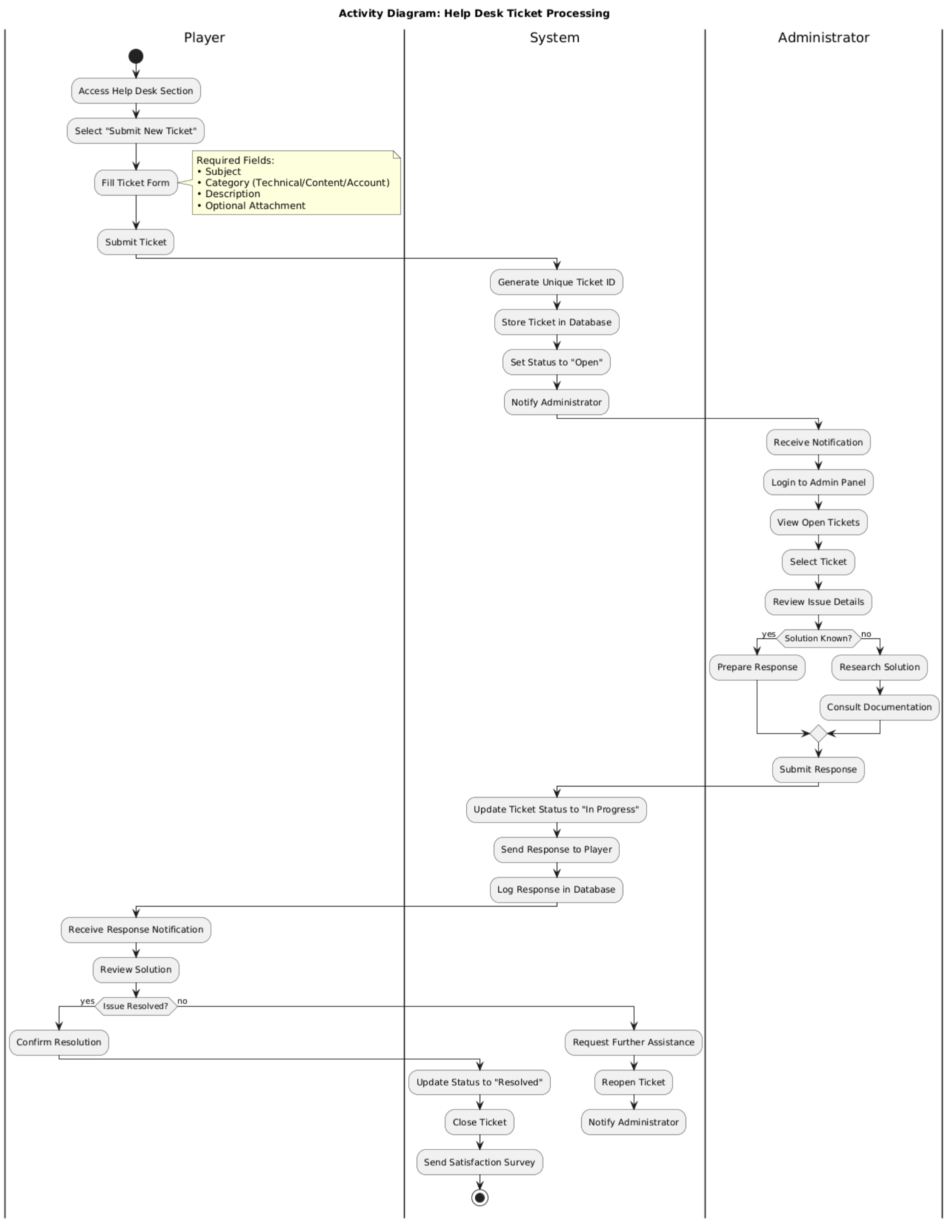


Figure 20: ACT-07: Help Desk Management

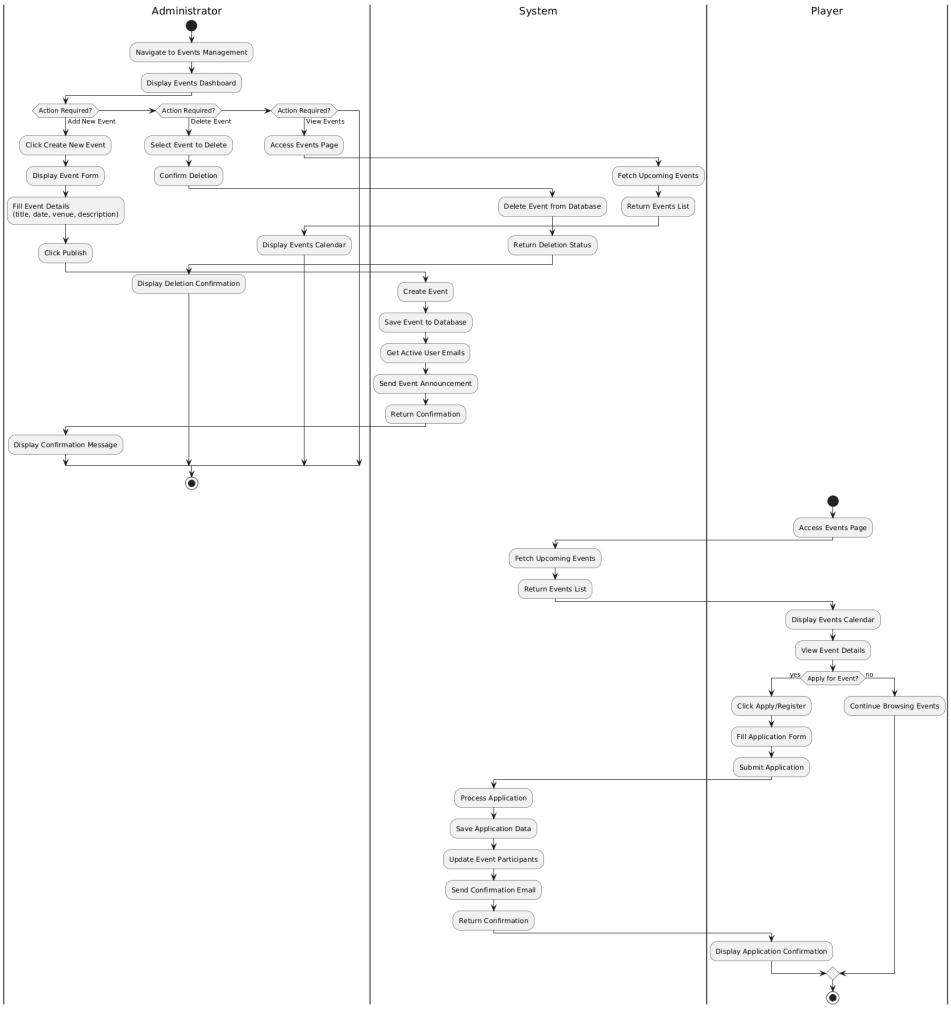


Figure 21: ACT-08: Ads and Events

4.5 Class Diagram

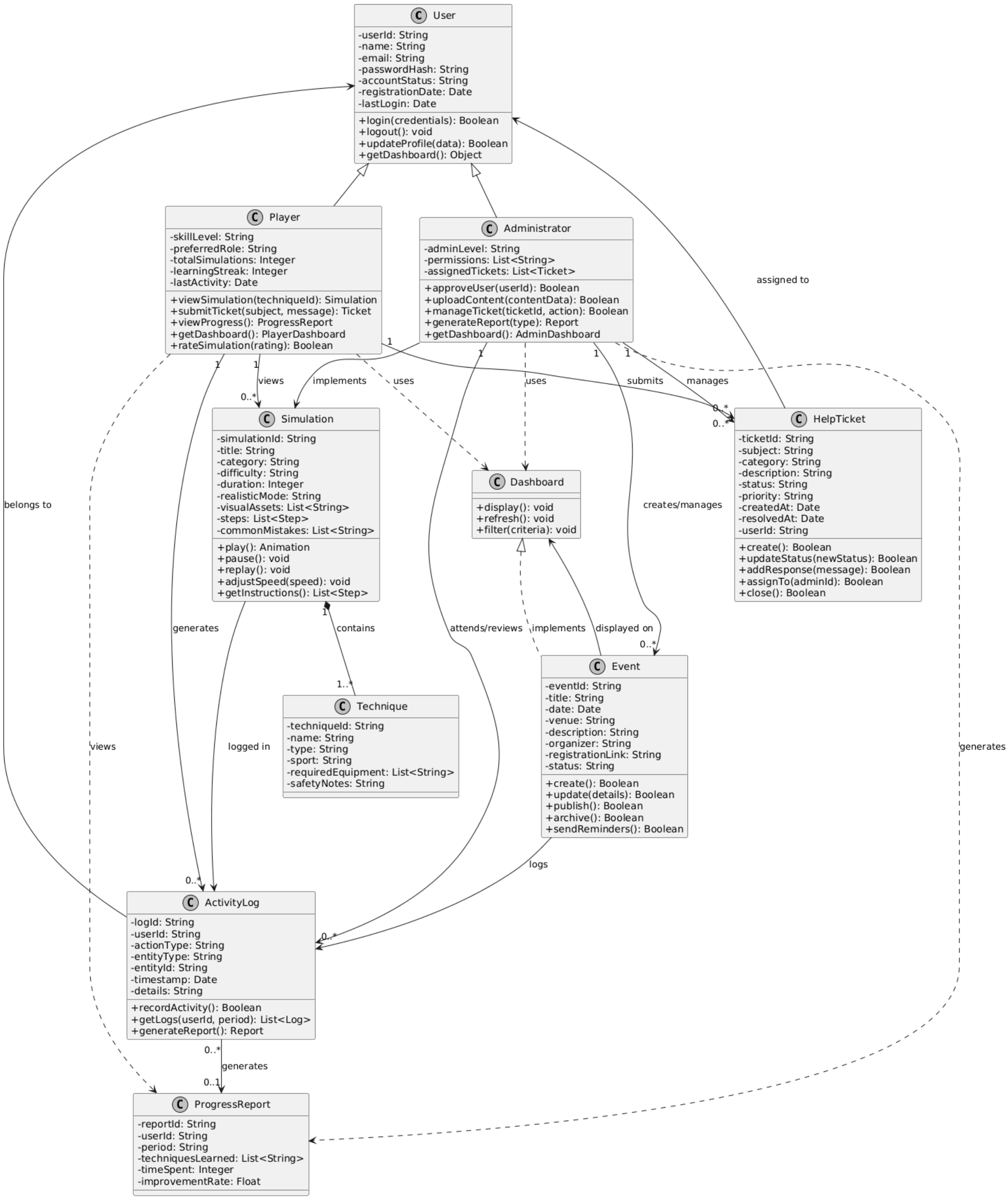


Figure 22: Class Diagram

4.6 Component Diagrams

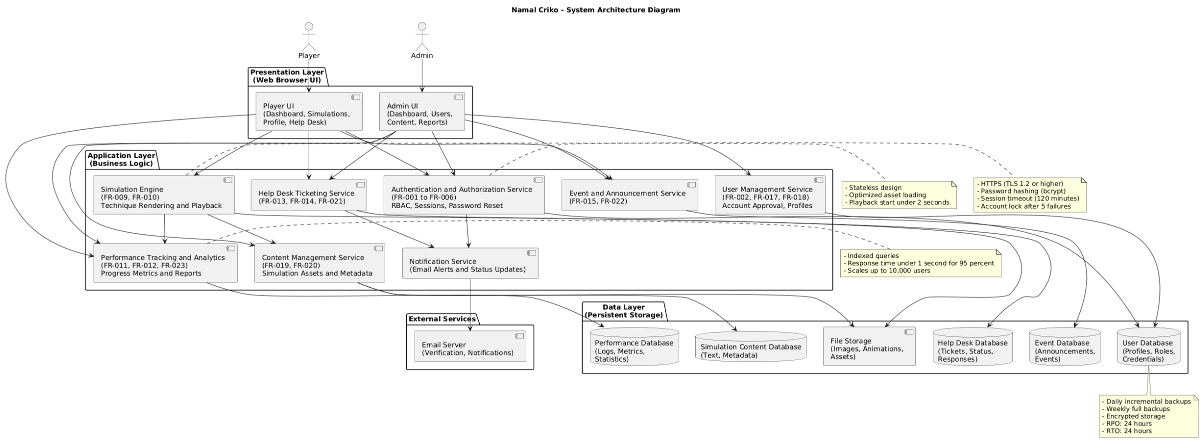


Figure 23: Component Diagram - System Architecture

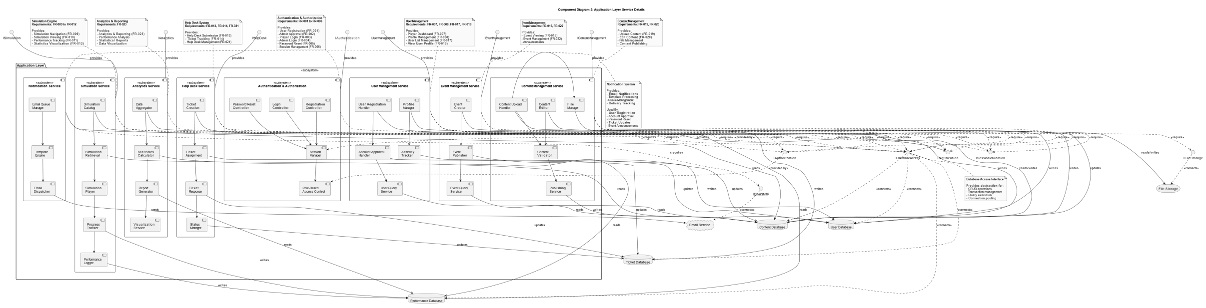


Figure 24: Component Diagram - Application Layer

4.7 Architecture Diagram

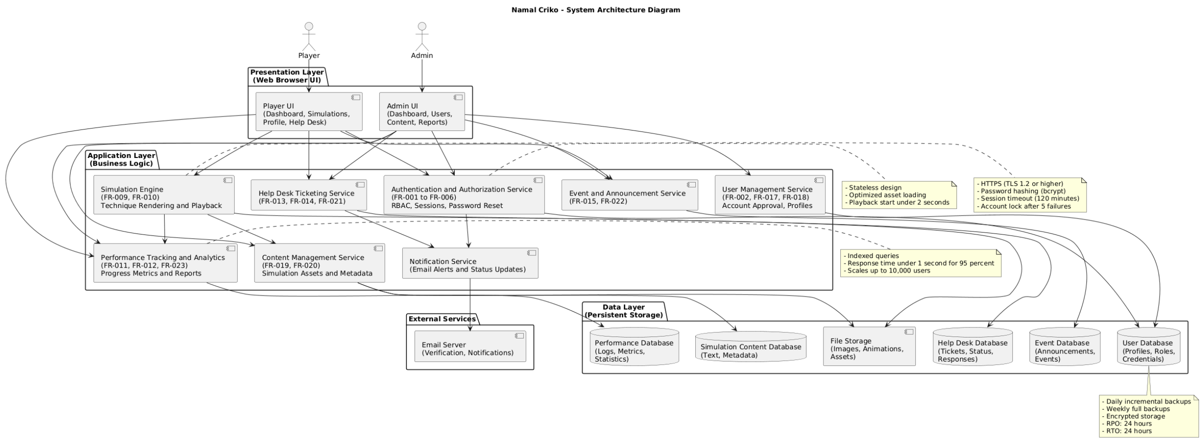


Figure 25: System Architecture

5 Requirements-Design Traceability

Legend: UC (Use Case), DFD (Data Flow Diagram), SD (Sequence Diagram), ACT (Activity Diagram), CD (Class Diagram), UI (User Interface), ARCH (Architecture)

5.1 Authentication & Authorization

Table 1: Traceability - Authentication

ID	Requirement	UC	DFD	SD	ACT	CD	UI	ARCH
FR-001	User Registration	UC-01	DFD-0	SD-01	ACT-02	CD-01	UI-01	ARCH-01
FR-002	Admin Approval	UC-01	DFD-1	SD-01	ACT-02	CD-01	UI-02	ARCH-01
FR-003	Player Login	UC-01	DFD-0	SD-02	—	CD-01	UI-03	ARCH-01
FR-004	Admin Login	UC-01	DFD-0	SD-02	ACT-03	CD-01	UI-04	ARCH-01
FR-005	Password Reset	UC-01	DFD-1	SD-03	—	CD-01	UI-05	ARCH-01
FR-006	Session Management	UC-01	DFD-0	SD-02	—	CD-01	UI-03	ARCH-01

5.2 Player Module

Table 2: Traceability - Player Module

ID	Requirement	UC	DFD	SD	ACT	CD	UI	ARCH
FR-007	Player Dashboard	UC-01	DFD-0	—	—	CD-01	UI-06	ARCH-01
FR-008	Profile Management	UC-01	DFD-1	—	ACT-01	CD-01	UI-07	ARCH-01
FR-009	Simulation Navigation	UC-01	DFD-0	SD-04	ACT-05	CD-01	UI-08	ARCH-01
FR-010	Simulation Viewing	UC-01	DFD-2	SD-04	ACT-05	CD-01	UI-09	ARCH-01
FR-011	Performance Tracking	UC-01	DFD-2	SD-04	ACT-04	CD-01	UI-06	ARCH-01
FR-012	Statistics Visualization	UC-01	DFD-1	—	ACT-04	CD-01	UI-10	ARCH-01
FR-013	Help Desk Submission	UC-01	DFD-1	SD-05	ACT-07	CD-01	UI-11	ARCH-01
FR-014	Ticket Tracking	UC-01	DFD-1	SD-06	ACT-07	CD-01	UI-12	ARCH-01
FR-015	Event Viewing	UC-01	DFD-1	SD-07	ACT-08	CD-01	UI-13	ARCH-01

5.3 Admin Module

Table 3: Traceability - Admin Module

ID	Requirement	UC	DFD	SD	ACT	CD	UI	ARCH
FR-016	Admin Dashboard	UC-01	DFD-0	—	ACT-03	CD-01	UI-14	ARCH-01
FR-017	User List Management	UC-01	DFD-1	SD-08	ACT-03	CD-01	UI-15	ARCH-01
FR-018	View User Profile	UC-01	DFD-1	SD-08	ACT-03	CD-01	UI-16	ARCH-01
FR-019	Upload Content	UC-01	DFD-2	SD-09	ACT-06	CD-01	UI-17	ARCH-01
FR-020	Edit Content	UC-01	DFD-2	SD-09	ACT-06	CD-01	UI-18	ARCH-01
FR-021	Help Desk Management	UC-01	DFD-2	SD-06	ACT-07	CD-01	UI-19	ARCH-01
FR-022	Event Management	UC-01	DFD-2	SD-07	ACT-08	CD-01	UI-20	ARCH-01
FR-023	Analytics & Reporting	UC-01	DFD-2	—	ACT-04	CD-01	UI-21	ARCH-01

6 Prototype Overview

6.1 Paper Prototype

Presented paper-based prototype in first RP meeting showing main interfaces and workflows. Feedback validated navigation structure.

6.2 Figma Prototype

Interactive prototype: https://figma.com/your-prototype-link

6.3 Screenshots

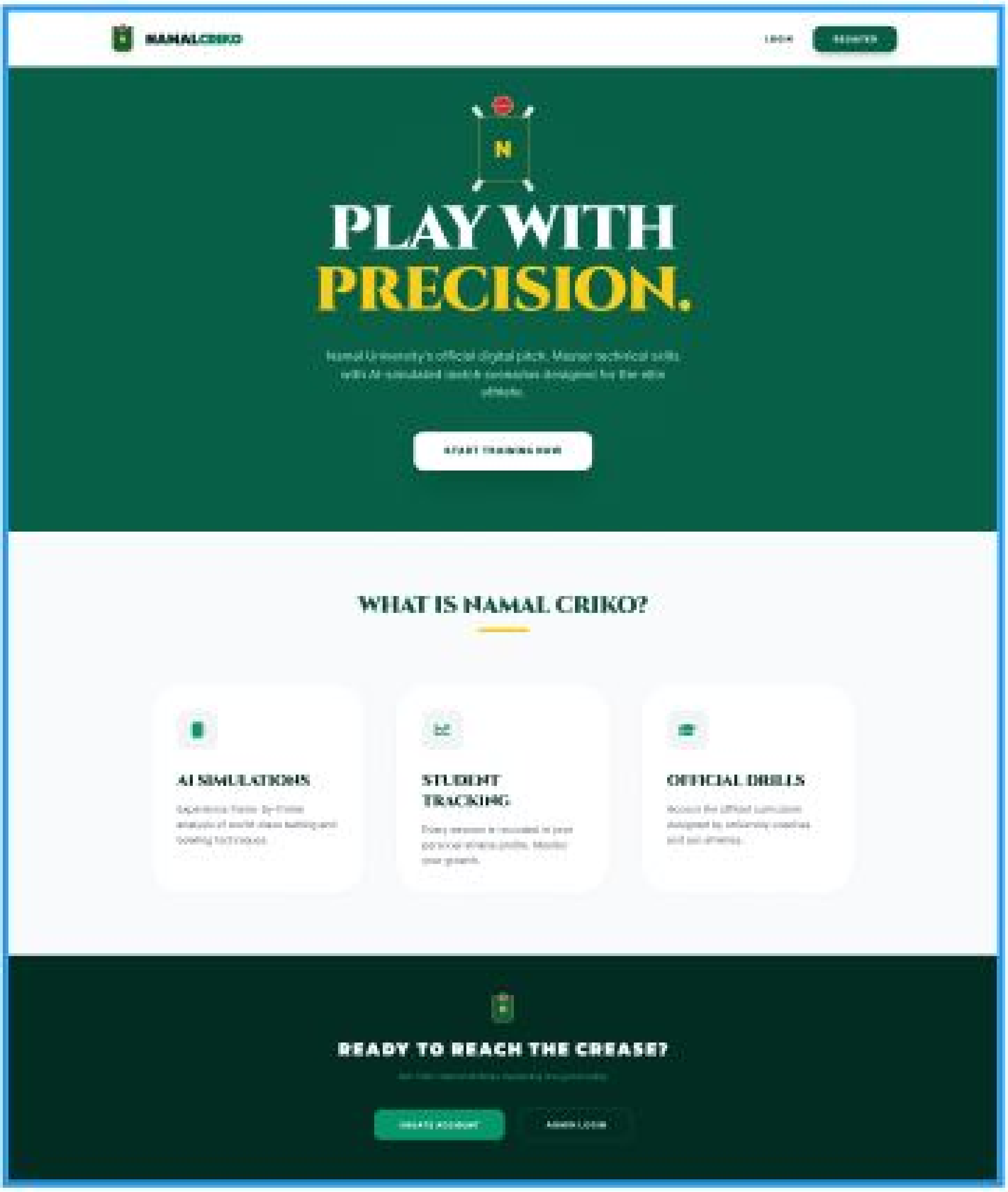


Figure 26: Landing Page

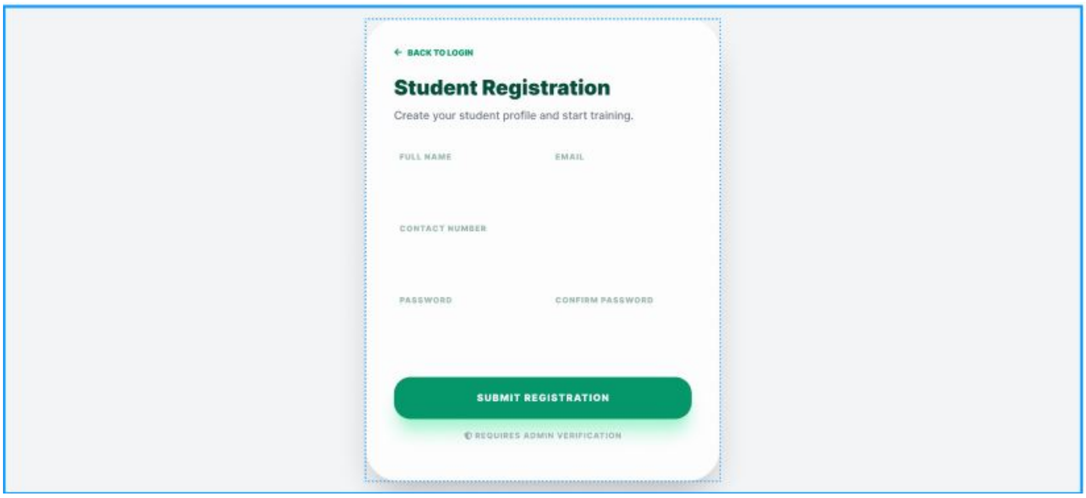


Figure 27: Registration Form

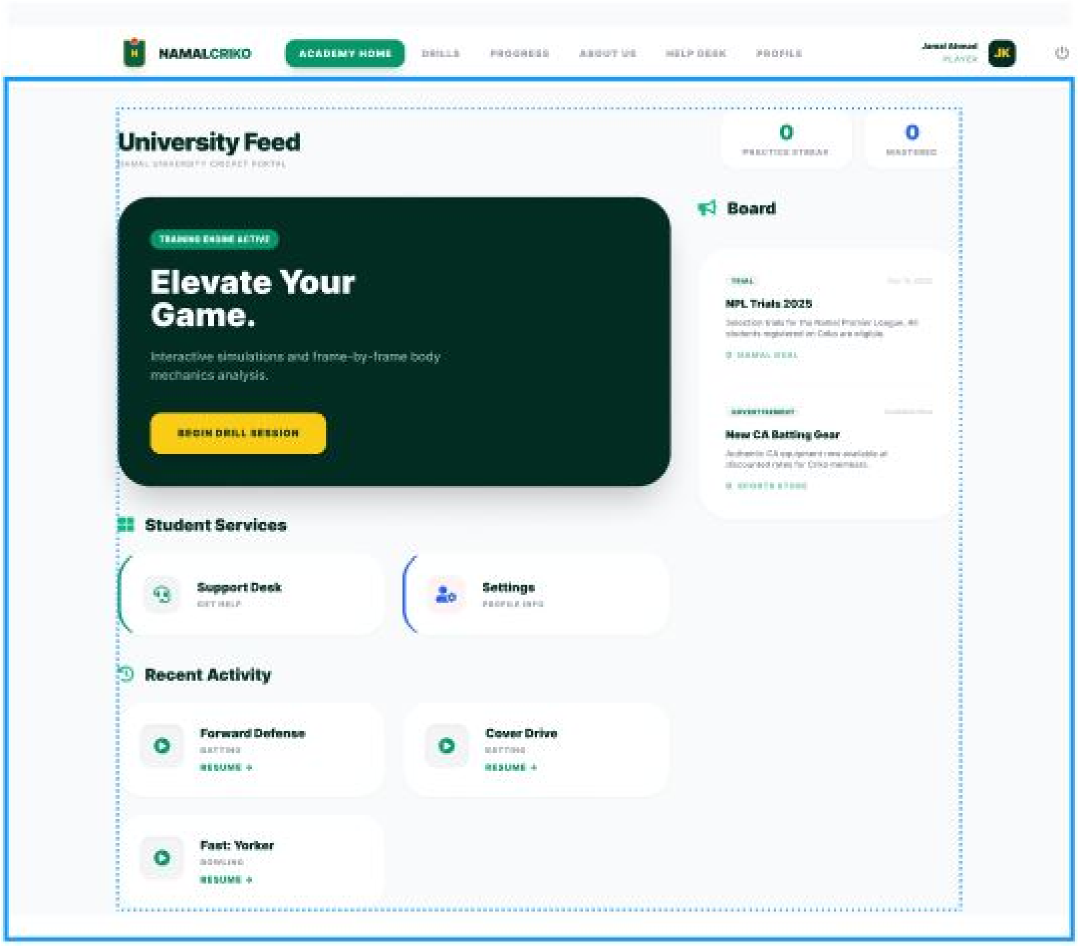


Figure 28: Player Dashboard

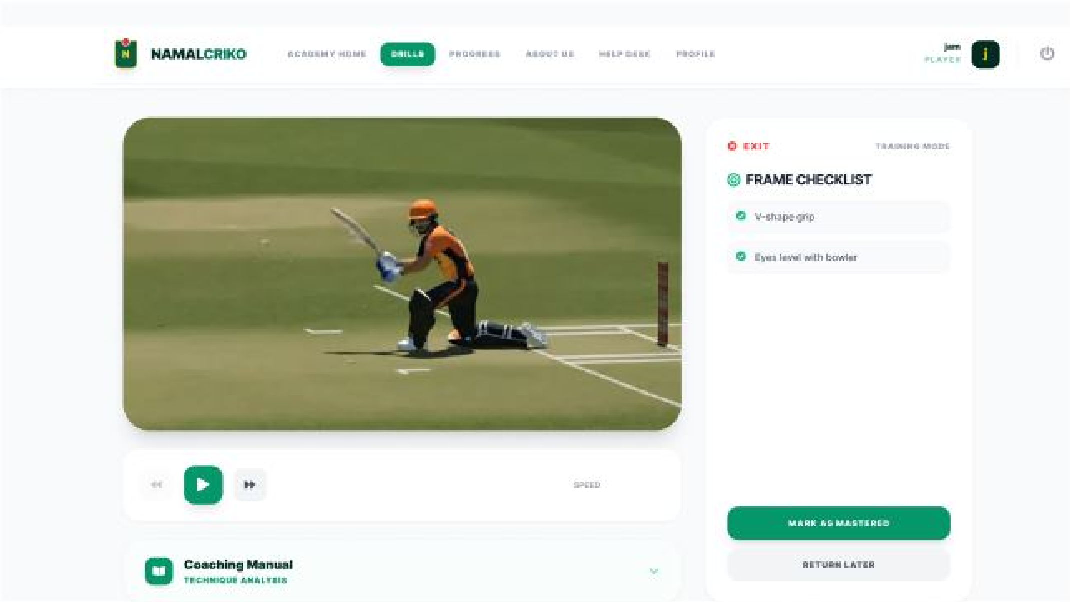


Figure 29: Simulation Viewer

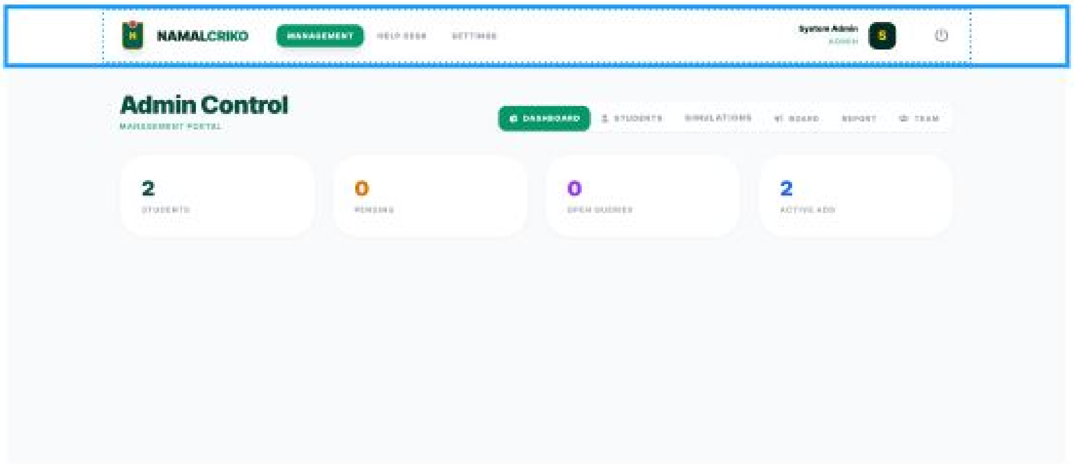


Figure 30: Admin Dashboard

6.4 Meeting Documentation

Meeting 1: Paper prototype presentation
Meeting 2: Figma prototype demonstration
Videos & Minutes: Available in GitHub repository

7 Project Links

GitHub Repository: https://github.com/MJamalAhmadKhan/Namal-Criko.git
Figma Prototype: https://www.figma.com/proto/iL4ZUEg8pJlOrQeghJq0av/Untitled